Predicting Ozone Nonattainment in US Counties with Supervised Learning

UE24CS352A — Machine Learning Mini-Project | Section 5E, Team 28
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Objective

This project investigates whether supervised machine-learning models can classify US county records according to ozone nonattainment status. The goal is to compare models using ozone measurement and air-quality summary features, with particular attention to identifying the positive (nonattainment) class.

Data and Methodology

The dataset combines 2025 EPA AirData daily ozone measurements (parameter code 44201) with county-level status labels derived from EPA Green Book information. Five input features are used: number of daily ozone records, mean daily ozone concentration, maximum daily ozone concentration, mean daily AQI, and mean observation percentage. The target is **nonattainment_label** (Class 0 or Class 1). The data are stratified into training and test portions. Logistic Regression, Decision Tree, and Random Forest are compared using five-fold stratified cross-validation on the training portion. Class weighting is used to address unequal class frequencies, and StandardScaler is included in the Logistic Regression pipeline.

Five-Fold Cross-Validation Results

Model	Accuracy	Precision	Recall	F1-score
Logistic Regression	0.781	0.476	0.770	0.588
Decision Tree	0.818	0.554	0.533	0.542
Random Forest	0.871	0.737	0.566	0.638

Results and Discussion

Random Forest achieved the highest mean accuracy (0.871), precision (0.737), and F1-score (0.638), making it the strongest overall model by these measures in the cross-validation comparison. Logistic Regression achieved the highest recall (0.770), indicating that it identified a larger share of positive cases; however, its lower precision (0.476) suggests more false-positive predictions. The Decision Tree produced intermediate accuracy but the lowest F1-score. These results show why accuracy alone is insufficient for an imbalanced classification task: precision and recall reveal different types of error.

Conclusion and Limitation

Based on the cross-validation results, Random Forest is the preferred overall model, while Logistic Regression may be useful when identifying as many positive cases as possible is the priority. This experiment classifies county records using measurements and labels associated with 2025; it does not forecast future nonattainment. Label construction and missing, blank, maintenance, or redesignation statuses can affect interpretation of the classes. Further validation on an untouched dataset and careful review of labeling rules are recommended before real-world use.

Metrics are saved five-fold cross-validation means from the project notebook. Precision, recall, and F1-score refer to the positive class (Class 1).